Nyx Haile

Software Engineer

xyn@mit.edu | LinkedIn | GitHub | Open to relocation

Education

Massachusetts Institute of Technology May 2026

B.S. Mathematics, Minor in Economics *Cambridge, MA*

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

Experience

Partial 2025 – Present

Cofounder & CEO *London*

- Co-founded Partial, a prediction market that turns any relational analysis between assets into a tradable position.
- Built Partial's Rust pricing engines, which assemble any contract from Boolean combinations of a few simpler contracts.

Weiss Laboratory & The SHED, MIT June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue *Cambridge, MA*

- Built the fluidics-design AI agent and wrote the master prompt for a proof-of-concept multi-agent LLM workflow that generates complex vascular STL designs from plain-text specifications; co-inventor on the MIT invention disclosure.
- Wrote a parametric vascular-network generator specification for the Cellvivo bioreactor, covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop.

Vana Dec 2022 – July 2023

AI Developer *Remote*

- Shipped an iMessage sticker app in Swift on the Vana API, generating personalised stickers from users' AI portraits.
- Engineered and iterated prompt templates for personalised AI-generated sticker portraits, tuning style and expression cues for identity retention and consistency.

MITERS, MIT Radio Society & SIPB Sept 2021 – Present

Keyholder (MITERS, Radio Society); SIPB member *Cambridge, MA*

- Contributed to arkOS, SIPB's self-hosted AI assistant, introducing a typed StateOutput model that its agent state machine returns and consumes.

Selected Projects

Repositories marked private are available on request.

Strand *(private)* | Rust, FoundationDB *Capability-authenticated issue tracker*

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Shipped a 16-crate, ~71k-line Rust workspace through v1.0.0 GA (10/10 verification gates) and v1.1.0, then a v2.0 governed-execution prerelease.

Forge *(private)* | Rust, systemd, RPC *Local-first agent control plane*

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Implemented an atomic file-write tool that records attribution for every write and reverts the file if recording fails; the default recorder fails closed.

howtowin.lol | Python, PyTorch, CUDA, Redis *Sequence model for post-game analysis*

- Designed a Dreamer-lineage RSSM sequence model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.
- Trained the model end to end on 12.8k games; at minute 15 it predicted the winner at 0.88 AUC on unseen matches and 0.84 on unseen players.

Sherlock | Rust, Dolt *Token-usage forensics for agent CLIs*

- Built a Rust CLI that ingests agent-CLI logs (Claude Code, Codex, Copilot), attributes token usage per session, and reports what burns tokens without adding value.
- Stored ingested usage in a Dolt versioned-SQL repository.

Technical Skills

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis **Web:** SvelteKit, React, Next.js